Table-Track

Pizza and Restaurant Management System — Project Documentation

1. Project Overview

Table-Track is a console-based (command-line) restaurant management system written in C. It is built for a small pizza/restaurant business and lets staff manage the menu, take customer orders, and keep a record of bills. The system supports two types of users — **Admin** and **User (staff)** — each with different permissions, so only admins can modify the menu while regular staff can mainly place orders and view items. It was developed in Code::Blocks (C language) to apply core concepts such as functions, structures, file handling, and modular program design in one realistic application.

2. How the Project Works

The program starts with a main menu: **Register**, **Login**, and **Exit**.

- **Register:** a new user chooses a username, password, and role (Admin or User); this is saved as a line in **users.txt** (duplicate usernames are rejected).
- **Login:** entered credentials are checked against **users.txt**; on success the user is logged in with the stored role.

After login, the menu shown depends on the role: the **Admin Menu** offers Add / View / Update / Delete / Search Menu Item, Place Order, and View Bill; the **User Menu** offers View, Search, Place Order, and View Bill.

All menu items live in **menu.txt** (ID, name, category, price, stock). Updating or deleting an item rewrites the file via a temporary file to keep data consistent. To place an order, the user enters an item ID and quantity; the program checks stock, calculates the bill (price × quantity), deducts stock, and appends the order to **orders.txt**. "View Bill" simply prints every line of this order history file.

3. Project Contents

The project is organized into separate modules, each handling one responsibility:

File	Responsibility
main.c	Program entry point; shows the main menu and routes to the Admin/User sub-menu.
Admin.c / .h	User accounts: register and login (reads/writes users.txt).
menu.c / Menu.h	Menu management: add/view/update/delete/search items (menu.txt); defines the MenuItem structure.
Order.c / .h	Order processing: place order, check stock, bill calculation, view bill history (orders.txt).
users.txt	Stores accounts: username, password, role.
menu.txt	Stores menu items: ID, name, category, price, stock.
orders.txt	Stores order history used for the bill view.

Core features included:

- User registration and login with role-based access (Admin / User).
- Add, view, update, delete, and search menu items with auto-incrementing IDs.
- Order placement with automatic stock checking, stock deduction, and bill calculation.
- Persistent order/bill history, saved to plain text files so no data is lost on exit.

4. Limitations

- **Plain-text password storage:** passwords are stored as plain text in users.txt, with no encryption or hashing, so this is not secure for real-world use.
- **No table/reservation tracking:** despite the project name "Table-Track", the current version does not actually manage restaurant tables or reservations — it only manages the menu, orders, and billing.
- **No edit/cancel for existing orders:** once an order is placed, it cannot be edited, cancelled, or removed from the order history.
- **Console-only interface:** there is no graphical user interface (GUI); everything is operated through a text-based menu, and data is stored in local .txt files instead of a proper database.

5. Future Work

- Add real table/reservation management (matching the project's name) — assigning orders to specific tables and tracking table availability.
- Encrypt or hash stored passwords instead of saving them as plain text.
- Link each order to the logged-in user's account so bills can be viewed per customer.
- Add the ability to edit or cancel a placed order.
- Replace text-file storage with a proper database (e.g. SQLite or MySQL) for better data handling and multi-user support.
- Add stronger input validation to prevent crashes or incorrect data from bad input.
- Build a simple GUI (or a basic web interface) to replace the console menu for easier use.
- Add printable/exportable bill receipts and daily/monthly sales reports for the admin.

Prepared as project documentation for the Table-Track Pizza and Restaurant Management System (C language, Code::Blocks).